Ying Cui

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EDUCATION

Doctor of Philosophy in Biostatistics — Aug 2017 - Apr 2023
Department of Biostatistics and Bioinformatics, Emory University, Atlanta, GA
Dissertation Advisor: Dr. Limin Peng
Dissertation Title: Flexible and Robust Methods for Evaluating Covariate Effects on Biomedical Outcomes

Bachelor of Science in Statistics — Aug 2013 - Jul 2017
Department of Mathematics, Nankai University, Tianjin, China
Thesis Advisor: Dr. Ximing Xu
Thesis Title: Composite Likelihood in Latent Class Modeling

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow — May 2023 - Present
Department of Biomedical Data Science, Stanford University, Stanford, CA
Supervisor: Dr. Lu Tian

Statistics Summer Intern — May 2021 - Aug 2021
Pfizer Inc.
Manager: Dr. Bo Huang

RESEARCH OVERVIEW

Research Interests: Precision Medicine, Statistical Machine Learning, Conformal Prediction, Quantile Regression, Causal Inference.

I develop statistical methods for precision health, with a focus on individualized variable selection, interpretable prediction, and uncertainty quantification. I study how to collect informative measurements efficiently and make reliable predictions and inferences from complex biomedical data. My research develops methods in quantile-based and semiparametric inference, reinforcement learning, and conformal prediction, motivated by clinical trials, drug development, and electronic health record studies.

SELECTED HONORS AND AWARDS

Scholarship Award, Biopharmaceutical Section, ASA — 2023
Student Paper Award 1st Place, Medical Device and Diagnostic Section, ASA — 2021
Distinguished Student Paper Award, ENAR International Biometric Society — 2021
Livingston Fellowship Award, Emory University — recognizes students with excellent research potential and strong ability to give impressive public presentations to showcase the PhD program — 2021
Outstanding Thesis Award and Outstanding Graduate, Nankai University — 2013-2017

RESEARCH GRANTS & PROPOSALS

Principal Investigator, NIH K99/R00 Pathway to Independence Award (Submitted Oct 2025; Pending)
National Institute of General Medical Sciences (NIGMS), NIH
Project: Efficient Drug Discovery Through Machine Learning Innovations

Collaborator, CANSSI's Collaborative Research Teams (CRT) program (2026-2029)
Canadian Statistical Sciences Institute (CANSSI)
Project: Collaborative Innovations in Statistical Learning for Next-Generation Drug Discovery
PI: Dr. Archer Y. Yang

Co-Investigator, Responsible AI for Safe and Equitable Health Seed Grants (2025-2026)
Stanford Institute for Human-Centered Artificial Intelligence (HAI) & Stanford Medicine
Project: Advancing Health Equity With Personalized Comorbidity Scores
PIs: Drs. Tina Hernandez-Boussard, and Lu Tian

PUBLICATIONS

†: co-first author

Statistical Methods-oriented Publications

1. **Cui Y**, Li AM, Charu V, Hwang YM, Hernandez-Boussard T and Tian L, 2026. Learning Interpretable Point-Based Clinical Risk Scores via Direct Optimization. *IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026)*. (Oral, 19% acceptance rate). Accepted.
2. **Cui Y**, Lai HJ, Peng L. Non-parametric Testing for Survival Data With Time-dependent Covariates. *Statistica Sinica*. Forthcoming.
3. **Cai, B**†, **Cui Y**†, Fu H, Lloyd-Jones D, Zhao L, Tian L. Dynamic Decision Making With Individualized Variable Selection. *Journal of the American Statistical Association: Theory and Methods*. 2026 Aug;121(554):1336–1348. doi: 10.1080/01621459.2025.2555060.
4. **Cui Y**, Peng L. Global Group Testing and Screening With Dynamic Effects. *Statistica Sinica*. 2026 Jul;36(3):1279-1300. doi: 10.5705/ss.202023.0285.
5. **Cui Y**, Jang JH, Mannino RG, Manatunga AK. Model-based clustering of multiple images incorporating covariates. *Statistical Methods in Medical Research*. 2026 May;35(5):945-961. doi: 10.1177/09622802251393631.
   – ASA Medical Device and Diagnostic Section's Student Paper Award 1st Place for JSM 2021.
6. **Cui Y**, Huang B, Dong G, Uozumi R, Tian L. An IPCW Adjusted Win Statistics Approach in Clinical Trials Incorporating Equivalence Margins to Define Ties. *Statistics in Medicine*. 2025 Jul;44(15-17):e70180. doi: 10.1002/sim.70180.
7. **Cui Y**†, Yu Q†, Manatunga A, Jang JH. A model-free framework for evaluating the reliability of a new device with multiple imperfect reference standards. *Biometrics*. 2025 Jan 7;81(1). doi: 10.1093/biomtc/ujaf025.
   – ASA Medical Device and Diagnostic Section's Student Paper Award 1st Place for JSM 2023.
8. **Cui Y**, Huang B, Mao L, Uno H, Wei LJ, Tian L. Inferences for the distribution of the duration of response in a comparative clinical study. *Clinical Trials*. 2024 Oct;21(5):541-552. doi: 10.1177/17407745241264188.
9. **Cui Y**, Peng L. Assessing dynamic covariate effects with survival data. *Lifetime Data Analysis*. 2022 Oct;28(4):675-699. doi: 10.1007/s10985-022-09571-7.
10. **Cui Y**, Peng L, Hu Y, Lai HJ. Assessing the Reproducibility of Microbiome Measurements Based on Concordance Correlation Coefficients. *Journal of the Royal Statistical Society: Series C*. 2021 Aug;70(4):1027-1048. doi: 10.1111/rssc.12497.
    – Distinguished Student Paper Award for the 2021 ENAR Spring Meeting.

11. **Guo Z**†, **Cui Y**†, Shi X, Birchler JA, Albizua I, Sherman SL, Qin ZS, Ji T. An empirical Bayesian approach for testing gene expression fold change and its application in detecting global dosage effects. *NAR Genomics and Bioinformatics*. 2020 Sep;2(3):lqaa072. doi: 10.1093/nargab/lqaa072.

12. Lu J†, Zhang Y†, **Cui Y**, Peng L, Chen Z. A novel Phase II single-arm hybrid design to minimize trial duration and enhance subsequent Phase III trial success rate. *Journal of Applied Statistics*. 2025;52(3):578-594. doi: 10.1080/02664763.2024.2382135.

Biomedical or Public Health Collaboration-oriented Publications

13. Kim H, **Cui Y**, and Gu T. 2026. Global quantile-based feature screening for high-dimensional biomedical prediction. *Machine Learning and Artificial Intelligence in Bioinformatics and Medical Informatics Workshop, IEEE BIBM 2026*. Accepted.

14. Daluwatumulle S, Jang JH, **Cui Y**, Manatunga AK. A Clustering Method for Hybrid Functional and Scalar Data. *Biometrical Journal*. Forthcoming.

15. Dong G, Gamalo-Siebers M, **Cui Y**, Huang B, Luo X, Tian L. Win statistics (win ratio, win odds, and net benefit): Noncollapsibility and standardization for randomized clinical trials. *Journal of Biopharmaceutical Statistics*. 2026 Feb 18;:1-17. doi: 10.1080/10543406.2026.2626058.

16. Chang C†, Farah H†, Gui H†, ... , **Cui Y**, ... , Shah N, and Daneshjou R. Red Teaming ChatGPT in Medicine: Real-world Insights on Model Behavior. *npj Digital Medicine*. 2025 Mar 7;8(1):149. doi: 10.1038/s41746-025-01542-0.

17. Dong G, **Cui Y**, Gamalo-Siebers M, Liao R, Liu D, Hoaglin DC, Lu Y. On approximate equality of Z-values of the statistical tests for win statistics (win ratio, win odds, and net benefit). *Journal of Biopharmaceutical Statistics*. 2025 May;35(3):457-464. doi: 10.1080/10543406.2024.2374857.

18. Wang D, Zheng S, **Cui Y**, He N, Chen T, Huang B. Adjusted win ratio using the inverse probability of treatment weighting. *Journal of Biopharmaceutical Statistics*. 2025 Jan 2;35(1):21-36. doi: 10.1080/10543406.2023.2275759.

19. **Wei Q**†, **Yao Z**†, **Cui Y**†, Wei B, Jin Z, Xu X. Evaluation of ChatGPT-generated medical responses: A systematic review and meta-analysis. *Journal of Biomedical Informatics*. 2024 Mar;151:104620. doi: 10.1016/j.jbi.2024.104620.

20. Meng G, Pan Y, Tang W, Zhang L, **Cui Y**, Schumacher FR, Wang M, Wang R, He S, Krischer J, Li Q, Feng H. imply: improving cell-type deconvolution accuracy using personalized reference profiles. *Genome Medicine*. 2024 Apr 29;16(1):65. doi: 10.1186/s13073-024-01338-z.

21. Wei Q, Wang Y, Yao Z, **Cui Y**, Wei B, Li T, Xu X. Evaluation of ChatGPT's performance in providing treatment recommendations for pediatric diseases. *Pediatric Discovery*. 2023 Dec;1(3):e42. doi: 10.1002/pdi3.42.

22. Wei Q, **Cui Y**, Wei B, Cheng Q, Xu X. Evaluating the performance of ChatGPT in differential diagnosis of neurodevelopmental disorders: A pediatricians-machine comparison. *Psychiatry Research*. 2023 Sep;327:115351. doi: 10.1016/j.psychres.2023.115351.

23. Dong G, Hoaglin DC, Huang B, **Cui Y**, Wang D, Cheng Y, Gamalo-Siebers M. The stratified win statistics (win ratio, win odds, and net benefit). *Pharmaceutical Statistics*. 2023 Jul-Aug;22(4):748-756. doi: 10.1002/pst.2293.

24. **Cui Y**, Dong G, Kuan PF, Huang B. Evidence synthesis analysis with prioritized benefit outcomes in oncology clinical trials. *Journal of Biopharmaceutical Statistics*. 2023 May 4;33(3):272-288. doi: 10.1080/10543406.2022.2141769.

25. Dong G, Huang B, Verbeeck J, **Cui Y**, Song J, Gamalo-Siebers M, Wang D, Hoaglin DC, Seifu Y, Mütze T, Kolassa J. Win statistics (win ratio, win odds, and net benefit) can complement one

another to show the strength of the treatment effect on time-to-event outcomes. *Pharmaceutical Statistics*. 2023 Jan;22(1):20-33. doi: 10.1002/pst.2251.

26. Patrawala M, **Cui Y**, Peng L, Fuleihan RL, Garabedian EK, Patel K, Guglani L. Pulmonary Disease Burden in Primary Immune Deficiency Disorders: Data from USIDNET Registry. *Journal of Clinical Immunology*. 2020 Feb;40(2):340-349. doi: 10.1007/s10875-019-00738-w.

Preprints and Manuscripts Under Review

- **Shen A**†, **Cui Y**†, van Eijk R, Tian L, Lu Y., 2026+. Doubly Robust Estimation for Longitudinal Hierarchical Composite Endpoints. (Manuscript available upon request)
- **Cui Y**, and Tian L., 2026+. Interval-Conditional Conformal Recalibration of Black-Box Prediction Intervals. (Manuscript available upon request)
- **Cui Y**, Peng L. and Tian L., 2026+. Learning When to Select or Average: An Adaptive Ensemble Prediction Framework. (Submitted)
- **Hwang YM**†, **Cui Y**†, Xu J, Pan T, Li R, Rice B, Tian L and Hernandez-Boussard T., 2026+. One Size Does Not Fit All: A Data-Driven Framework for Personalized Comorbidity Scoring. medRxiv, pp.2026-06. (Submitted)
- **Wei Q**†, **Cui Y**†, **Ding M**†, Wang Y, Xiang L, Yao Z, Chen C, Long Y, Jin Z and Xu X., 2026+. Performance Evaluation of Lightweight Open-source Large Language Models in Pediatric Consultations: A Comparative Analysis. arXiv preprint arXiv:2407.15862. (Submitted)
- **Cui Y**, Li E, Uno H, Tian L., 2026+. A New Framework for Assessing Predictive Distributions with Censored Outcome. (Submitted)
- Shen H, Tian L, Zhang C, **Cui Y**, Liu M, Zhang W, Chen Z., 2026+. Targeted Learning for Treatment Effects on Time Lost Under Competing Risks in Electronic Health Records. (Submitted)
- Paydarfar D, **Cui Y**, Tian L, Solomon SD, Claggett B, Wei LJ., 2026+. Quantifying the Totality of Treatment Effect Across Multiple Endpoints in Clinical Studies. *Journal of the American Statistical Association*. (Invited resubmission)

Conference Abstracts

- **Hwang YM**†, **Cui Y**†, Xu J, Pan T, Li R, Rice B, Tian L and Hernandez-Boussard T, 2026, May. Data-Driven Personalized Comorbidity Framework: A Population-Stratified Approach. In 2026 Annual Research Meeting. AcademyHealth.
- **Cui Y**, Peng L, Hu Y, Sudakaran S, Murali S, and Lai H, 2021. 483: Statistical analysis for assessing reproducibility of gut microbiome compositional measurements in young children with cystic fibrosis. *Journal of Cystic Fibrosis*. 20, pp.S228-S229.

TEACHING AND MENTORING

Stanford University
Co-Instructor

- Short Course "On RMST in survival analysis", The 8th Stat4Onc Annual Symposium, May, 2025

Lab Instructor/Teaching Assistant

- BIODS 250 (Clinical Trial Design in the Age of Precision Medicine): Winter 2024.

Mentor for Junior Researcher

- Haoning Shen, Stanford University Aug 2026 - Present

- Arlina Shen, Stanford University — May 2026 - Present
- Albert Li, Stanford University — Mar 2026 - Present
- Emma Li, Stanford University — Aug 2025 - Present
- Hyun Kim, Columbia University — May 2025 - Jul 2026
- Vaibhavi N. Kulkarni, Stanford University — Jan 2024 - Apr 2024

Emory University
Teaching Assistant

- BIOS 710 (Probability Theory II, PhD core course): Fall 2018-2021
- BIOS 711 (Statistical Inference II, PhD core course): Spring 2019-2021

Doctoral Qualifying Exam Mentor

- Organized and led review sessions for PhD theory and methods qualifying exams during 2019–2022.

Mentor for Junior Researcher

- Qi Yu, Emory University — Jan 2022 - Aug 2023

INVITED & CONFERENCE TALKS

Invited talk, Joint Statistical Meetings 2027, Chicago, IL, USA.	2027
Invited talk, Duke Industry Statistics Symposium 2027, Cary, NC, USA.	2027
Invited talk, AMIA 2026 Annual Symposium, Dallas, TX, USA.	2026
Invited talk, The 7th ICSA-Canada Chapter Symposium, Montreal, Canada.	2026
Invited talk, Duke Industry Statistics Symposium 2026, Cary, NC, USA.	2026
Invited seminar talk, Department of Epidemiology and Biostatistics at IU Bloomington, Virtual.	2026
Invited talk, The 41st Korean International Statistical Society Webinar, Virtual.	2026
Invited talk, Joint Statistical Meetings 2025, Nashville, TN, USA.	2025
Invited talk, The fourth Lifetime Data Science (LiDS) Conference, Brooklyn, New York, USA.	2025
Invited talk, ICSA 2024 Applied Statistics Symposium, Nashville, TN, USA.	2024
Invited talk, Duke Industry Statistics Symposium 2024, Durham, NC, USA.	2024
Invited seminar talk, Division of Biostatistics, University of California, Berkeley, CA, USA.	2024
Invited talk, ICSA 2023 Applied Statistics Symposium, Ann Arbor, MI, USA.	2023
Invited talk, 2023 Lifetime Data Science Conference, Raleigh, NC, USA.	2023
Contributed talk, Eastern North American Region 2023 Spring Meeting, Nashville, TN, USA.	2023
Contributed talk, Joint Statistical Meeting 2022, Washington, DC, USA.	2022
Contributed talk, Eastern North American Region 2022 Spring Meeting, Houston, TX, USA.	2022
Student research talk, Explore Emory - student research session, Atlanta, GA, USA.	2022
Paper award talk, Joint Statistical Meetings 2021, Virtual.	2021
Paper award talk, Eastern North American Region 2021 Spring Meeting, Virtual.	2021

PROFESSIONAL SERVICE

Editorial Roles

- *Associate Editor*, Journal of Biopharmaceutical Statistics

Peer Review

- *Referee*: Annals of Applied Statistics, Biometrics, BMC Medical Research Methodology, BMJ Open, Computational Statistics & Data Analysis, F1000Research, IEEE BIBM 2023, JCO Precision Oncology, Journal of Big Data, Journal of Biopharmaceutical Statistics, Journal of Causal Inference, Journal of Machine Learning Research, Journal of Statistical Theory and Practice,

Journal of the American Statistical Association, Journal of the Royal Statistical Society: Series C, Lifetime Data Analysis, npj Digital Medicine, Scandinavian Journal of Statistics, Scientific Reports, STAI-X 2026, Statistical Methods in Medical Research, Statistics in Biopharmaceutical Research, Statistics in Biosciences, Statistics in Medicine.

Conference Organization and Service

- *Session Co-organizer*, Joint Statistical Meetings 2027, August 8–12, 2027, Chicago, IL, USA.
- *Session Chair*, Joint Statistical Meetings 2025, August 2–7, 2025, Nashville, TN, USA
- *Session Organizer & Chair*, Fourth Lifetime Data Science (LiDS) Conference, May 28–30, 2025, Brooklyn, NY, USA
- *Session Co-organizer & Chair*, Sixth ICSA–Canada Chapter Symposium, June 7–9, 2024, Niagara Falls, Canada
- *Session Chair*, ENAR 2023 Spring Meeting, March 19–22, 2023, Nashville, TN, USA

SOFTWARE

Foldseq: An R package for an empirical Bayesian method to detect log-fold change. Available on GitHub at `https://github.com/cuiyingbeicheng/Foldseq`.

agree.comp: An R package for assessing the reproducibility of microbiome measurements based on concordance correlation coefficients. Available at `https://web1.sph.emory.edu/users/lpeng/Rpackage.html`.

WINS: An R package to calculate the win statistics for prioritized multiple endpoints, plot the win statistics and win proportions over study time, and simulate datasets with dependent endpoints. Available on CRAN at `https://CRAN.R-project.org/package=WINS`.

survRDOR: An R package to perform inference on the distribution of the duration of response. Available on GitHub at `https://github.com/cuiyingbeicheng/survRDOR`.

CERTIFICATIONS

Cosmos Data Model for Data Architects Certification, Cosmos Academy — Certified in EHR data modeling and line-level access (SQL) within Epic Cosmos 2025

SAS Certified Base Programmer for SAS 9, SAS 2017

REFERENCES

Dr. Limin Peng
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Rollins School of Public Health, Emory University
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Dr. Amita Manatunga
Donna Jean Brogan Professor of Biostatistics
Rollins School of Public Health, Emory University
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Dr. Lu Tian
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School of Medicine, Stanford University
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Dr. Lexin Li
Professor of Biostatistics
School of Public Health, University of California, Berkeley
Email: lexinli@berkeley.edu

Dr. Ying Lu
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